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Original Research

Evaluation of a PDA-based Dietary Assessment and Intervention Program: A Randomized Controlled Trial

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Objective: To evaluate the capability of DietMatePro, a PDA-based dietary assessment program, to monitor dietary intake and to improve adherence to a dietary regimen.

Design: Randomized controlled trial.

Subjects. Overweight and obese (Body Mass Index (BMI) 25–40) participants without dietary restrictions.

Intervention: Participants ($n = 174$) were randomized to record usual dietary intake using either DietMatePro or a paper food diary for one week to compare concordance with 24-hr recall. At the week 1 visit, participants were individually counseled to follow the diet recommendations of the Ornish Prevention Diet for three weeks and continue monitoring food intake using the assigned method to estimate adherence to the intervention by monitoring condition.

Outcome Measures: Spearman correlations between week 1 24-hr recall and the assigned recording method were compared to assess validity. Mean pre-post changes in intake measured by 24-hr recall were compared according to monitoring condition to measure adherence to the Ornish diet.

Results: Correlations of energy and nutrient values reported on the food label ranged from 0.41 to 0.71 for the DietMatePro condition versus 0.63 to 0.83 for the paper-based diary. Diet adherence was higher among DietMatePro (43%) compared to the paper diary (28%) group ($p = 0.039$).

Conclusions/Applications: DietMatePro does not appear to produce more valid data than paper-based approaches. DietMatePro may improve adherence to dietary regimens compared to paper-based methods.

INTRODUCTION

Lack of adherence to dietary regimens is a pervasive problem in both clinical nutrition and nutrition research. In research, lack of adherence to assigned interventions yields an intention-to-treat analysis biased towards null findings. One effective strategy for increasing adherence is self-monitoring of dietary intake [1–3]. Though self-monitoring is an indirect marker of motivation, evidence suggests that monitoring intake also directly leads to behavior change [4,5]. Traditional self-monitoring involves writing down everything one eats and drinks, and often includes referencing tables to calculate energy and/or nutrient goals [6]. This process is time-consuming and inconvenient for patients and study participants, especially when

eating away from home, which can result in retrospective completion of paper-based food diaries.

Paper-based food records are also cumbersome when used in clinical practice or nutrition research because they require researchers to enter foods and portions into a nutrient database for tabulation. As prospective measures of food intake, however, they may be one of the better measures of dietary assessment compared to 24-hr recalls or food frequency questionnaires [7,8], though no single measure of energy and nutrient intake can be considered acceptably adequate [9]. When completed during or immediately after a meal, food diaries minimize memory recall errors. Recall methods such as 24-hr recall, however, are less reactive than prospective recording which tends to influence food consumption.

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All authors were employed at PICS at the time of the study, and PICS is the developer of DietMatePro. This study was funded by NINR Grant# R44NR008443.

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Technological advances have facilitated the process of calculating energy and nutrient intakes from food records. Web-based programs allow end-users to enter food intake and receive feedback regarding energy and nutrient intake. A more portable option is Personal Digital Assistant (PDA)-based programs that provide tracking of intake against daily energy and nutrient goals [10]. Dietary data can be automatically and immediately uploaded for the researcher or clinician to analyze.

PDA-based programs such as DietMatePro (PICS, Version 1, 2003, Reston, VA,) offer tailored feedback and features designed to improve adherence to dietary regimens. DietMatePro provides individualized intake recommendations, comparison of actual intake with targeted intake, reminders to record meals, and meal plans and recipes that meet diet specifications. However, it is unclear whether PDA-based programs are feasible alternatives to written food records, given that PDA programs require additional participant training and shift the burden of data entry from the health care professional to the end-user.

Researchers have examined both the validity and utility of using PDA-based programs for dietary research [11,12]. Uncontrolled studies using PDA-based dietary assessment programs to improve adherence to dietary regimens has provided data to support the feasibility of the approach in diverse populations and over periods of up to six months [13–16]. To the best of our knowledge, this is the first report in the literature of a randomized, controlled trial comparing the utility of electronic versus paper-based food diaries in a large sample.

The purpose of this 4-week randomized, controlled trial was to evaluate the effectiveness of DietMatePro. Two primary aims were to 1) compare concordance of reported energy and nutrient intakes obtained from DietMatePro and paper food diaries with 24-hr recall; and 2) compare adherence to a strict dietary regimen among users randomized to use DietMatePro versus a paper-based food record.

MATERIALS AND METHODS

Participants ($n = 174$) were recruited from November 2003 to March 2004 from print ads placed in the health section of the *Washington Post* seeking volunteers for a four week diet study of food intake recording methods (Fig. 1). Eligible participants were overweight (BMI 25–40) adults who were computer literate (reported using computers at least 3 times per week) and able to read the standard displays of a Palm Zire 21. Physician approval to follow a very low fat, vegetarian diet was also required to participate in the dietary intervention period. Participants reporting following a special diet for any health condition were excluded from the study.

A security deposit of \$50 was obtained from each participant at the beginning of the study and was returned when the study materials were returned. Participants received \$100 for completing the study. The PICS Institutional Review Board

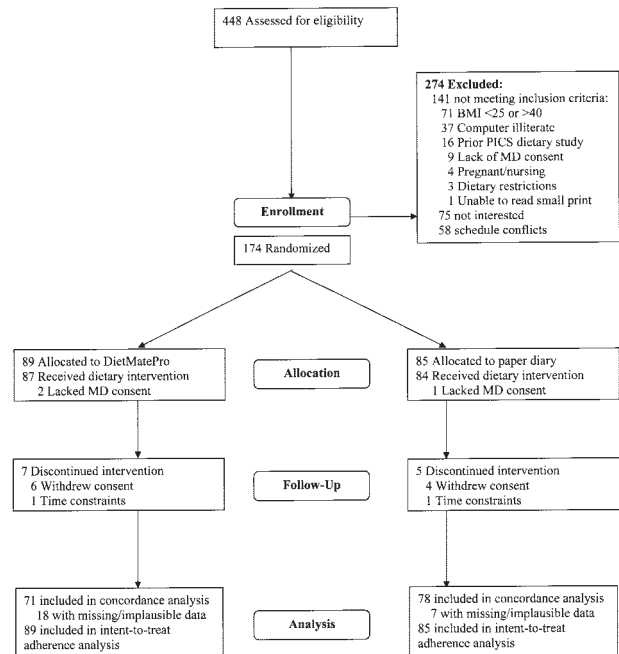


Fig. 1. Study Participant Flow.

approved the study, and all participants provided written informed consent.

Study Procedures

Potential participants called the research office in response to newspaper ads and were screened for eligibility. A research assistant next contacted the potential participant's physician to obtain approval to follow the Ornish Prevention Diet, a very-low fat, vegetarian diet. The Ornish diet was selected because it is diet with considerable research support [17] but is also a restrictive diet that is difficult for most to follow [18].

At the baseline visit, participants completed the demographic questionnaire and a trained research assistant measured weight using a Detecto triple beam physician's scale and waist circumference using a tension-controlled tape measure. Participants were then randomly assigned to receive either the DietMatePro program or the paper-based food diary as their food recording method based on a randomization table generated by the first author. Research staff that made assignments based on this randomization table were not masked to randomization at the baseline visit but eligibility determinations were made based on phone screenings prior to the baseline visit.

Participants were provided instructions for their assigned recording method and asked to record intake of all food and drink consumed throughout the entire 4-week study period. All participants were provided with a food portion education pamphlet to assist in determining appropriate food portion sizes for recording using either assessment method. Study participants were instructed not to make any dietary changes during the initial week they were in the study.

Those in the paper-diary condition received a 31 page, 5.5' by 8.5' food diary booklet that was based on a review of other food diaries used by dietary researchers. Each page had spaces for entries of all foods eaten at breakfast, lunch, dinner, and snacks for each day and the estimated portions for each food eaten. Participants were instructed to record an amount and detailed description of all food and drink consumed as close to the eating occasion as possible throughout the 4-week study period.

Those in the DietMatePro condition received a Palm Zire 21 loaded with the DietMatePro program. The participant could access the program at any time from the DietMatePro icon but was also prompted by an auditory alert to record meals at the times the user was most likely to eat these meals each day. After selecting the meal to record, foods from version 15 of the USDA database, augmented with common packaged and restaurant foods, could be selected using either a search function or selecting within food category lists. After each selection, the program prompted the user to enter the unit of measure and the amount of each food selected. Foods from meals could be saved and later modified and reused for subsequent recording if similar to a prior meal, thus reducing food entry time. Foods not listed in the program could be entered individually along with all of the information on the nutrition label. This database included a time and date stamp for each recording, the time and date of the meal as recorded by the participant, the foods eaten at each meal, their portion size, and the nutrient values associated with this dietary intake. During the first week of DietMatePro use, participants recorded food amounts and the program displayed food amounts without any energy or nutrient feedback.

At the week 1 visit, the participant returned the recording method provided and the data were either uploaded (DietMatePro) or collected by the research assistant and later entered into the ESHA Food Processor SQL 9.1.2 (ESHA, Version 9.1.2, Salem, OR) to determine nutrient values. A research assistant reviewed paper-based food diaries to insure complete entries and appropriate portion estimations. Research assistants were aware of the participant's randomization assignment during the assessment. Participants completed subjective ratings of the recording method used.

The research assistant next conducted a 24-hr dietary recall with the participant. The week 1 24-hr recall served as the criterion measure for assessing concurrent validity while the purpose of the week 4 24-hr recall was to measure adherence to the dietary intervention. Prior to study initiation, a registered dietitian trained the research assistants to conduct a multiple pass 24-hr recall, using the USDA five-step multiple-pass method as a framework [19,20]. Following the training session, each research assistant conducted a mock 24-hr recall interview to insure consistency among interviewers. Food models were used to assist in portion-size estimation of consumed foods.

A research assistant followed a script to describe the very low fat diet to each participant. Each participant was provided

with an individualized target calorie level based on the Harris-Benedict calculation using NIH guidelines for weight loss [21], a fat goal of 10–15% of calories, saturated fat <7% calories, and cholesterol less than 200 mg. All participants were provided with *Eat More, Weigh Less* by Dean Ornish [22] and *Fat Free and Easy* by Jennifer Raymond [23]. Participants in the DietMatePro condition received a version of the program that displayed personalized target values for energy based on the Harris-Benedict calculation using NIH guidelines for weight loss [21] as well as fat, saturated fat, and cholesterol goals based on Ornish Prevention Diet recommendations. Additional DietMatePro program features to assist in adhering to the dietary regimen included feedback of comparisons between actual and target intake by meal and by day as well as recipes and meal plans consistent with the assigned diet. Research assistants instructed DietMatePro participants on these additional features.

All participants were asked to continue recording all foods and drinks consumed until they returned for their week 4 visits. At the end of the fourth week, participants returned their assigned recording method. Data collection at the final visit included a 24-hr recall, self-report of recording practices, weight, and waist measurements.

Data Entry

Two registered dietitians and one research assistant having previous experience with dietary research entered 24-hr recall and food diary information into ESHA Food Processor SQL (Version 9.1.2, Salem, OR). For quality control, 10% of the data were independently verified. Correlations between the two entries were high for energy, fat, saturated fat, and cholesterol, ranging from 0.91 to 0.97. Both the ESHA Food Processor and the DietMatePro system rely on the USDA Nutrient Database SR, and database differences were negligible [11]. Questionnaires were scanned using teleforms and data were verified by a research assistant.

Data Analysis

Independent t-tests for continuous variables and chi-squared tests for categorical variables were used to compare baseline values between conditions and between study participants who completed the study versus those who discontinued. Dietary variables with non-normal distributions were analyzed using non-parametric Wilcoxon rank sum tests.

To evaluate dietary monitoring capabilities of DietMatePro, week 1 median and inter-quartile ranges (25th and 75th percentile) reported by assigned recording method were calculated. To assess statistical significance between conditions, values were ln-transformed to improve normality and compared using independent t-tests. Concordance between week 1 24-hr recall and assigned recording method data were compared by Spearman correlations and compared using independent t-tests after Fisher transformation.

Dietary monitoring adherence via the DietMatePro diary and the paper-based food record was determined by calculating the percentage of days with plausible energy intake ranges (500 to 3500 calories for women and 800 to 4,000 calories for men) [24] by sampling the last three days, including weekend days if within this three day period, of week 1 and week 4. We sampled only six days out of the four-week intervention period due to the resources required to enter four weeks of paper-based food entries for each of the diary participants.

Reporting plausible energy intake ranges was used as a proxy for recording a “complete” day of intake and inclusion in subsequent analyses. Adherence to the prescribed diet regimen was analyzed using both the diary and 24-hr recall data. The percentage of recorded days during week 4 for each dietary monitoring system in which the participant remained at or under the total calories, fat, saturated fat, and cholesterol prescribed was calculated. Sensitivity analysis using worst and best case scenarios for missing data were performed.

Pre-post intervention changes in energy and nutrient intake were calculated by subtracting week 4 24-hr recall estimates from week 1 estimates. With 160 participants, the study provided over 90% power to detect a 10g (Standard Deviation (SD) 25g) difference in daily total fat intake with a 2-sided alpha level of 0.05 (PS Power and Sample Size, Version 2.1.3, 2003, Nashville, TN). We used SPSS version 12 (SPSS Inc, Chicago, IL) for statistical analyses, and the level of statistical significance was set at $p < 0.05$.

RESULTS

Preliminary Analysis

Despite randomization, mean BMI was 2 units lower in the DietMatePro condition than the paper diary condition ($p = 0.02$). There were no other significant differences between groups by randomization assignment (Table 1). At week 1, median energy intake reported by 24-hr recall was 1852 (IQR = 1437 to 2297) calories in the DietMatePro compared to 1605 (IQR = 1350 to 2015) in the paper diary group. This difference in intake between conditions was also observed comparing week 1 diary data. For the three main nutrients targeted in the intervention (fat, saturated fat, and cholesterol), reported intake was between 17% and 33% greater in the DietMatePro compared to the paper diary condition (Table 2).

Compared to the 159 participants who completed the intervention, the 15 participants who dropped out had a higher mean BMI (35.05 versus 30.63, 95% CI for difference 1.64 to 7.20) and waist circumference (38.06 vs. 35.72 in), 95% CI for difference 0.73 to 3.96 in). Completers did not differ from drop-outs in age, gender, level of education, employment status, or race.

Table 1. Baseline Characteristics of Study Participants by Monitoring Condition

	DietMatePro (n = 89)	Paper Diary (n = 85)
	Mean $\pm$ SD* or %	Mean $\pm$ SD or %
Age (y)	52 $\pm$ 12	54 $\pm$ 10
Female (%)	83	77
Education (y)	17 (2)	17 (2)
Married (%)	64	58
Employed full-time (%)	39	52
Caucasian (%)	85	83.3
Black (%)	10	16.7
Asian (%)	5	0
BMI (kg/m ²)**	29.9 $\pm$ 4.3	32.1 $\pm$ 4.4
Waist (in)	36.2 $\pm$ 4.9	38.6 $\pm$ 5.6

* SD = standard deviation, y = years, BMI = Body Mass Index, kg = kilograms, m = meters, in = inches.

** $p = 0.02$, unpaired t-test.

Concordance of Week 1 Paper Based and DietMatePro Diaries to Week 1 24-hr Recall

Despite similar summary measures of intake reported by both paper-based and DietMatePro food diaries, Spearman correlation coefficients measuring the association between 24-hr recall and the corresponding time period for the food diary assessed at week 1 were higher for the paper-based diary compared to DietMatePro (Table 2). The week 1 median difference between the paper diary and the recall was only 5 calories (IQR = -365 to 204) compared to 137 calories (IQR = -487 to 371) for the DietMatePro condition, but this difference was not statistically significant. For both conditions, the diary reflected higher energy and nutrient intake compared to the 24-hr recall.

Adherence to Dietary Monitoring between Conditions

Plausible caloric intake was defined as 500 to 3500 calories for women and 800 to 4,000 calories for men [24]. For the week 1 concordance analysis, data with missing or implausible energy intakes were reported for a larger proportion of DietMatePro participants (20%) compared to those assigned to the paper-based diary condition (8%) ($p = 0.03$). Over the 6 day sampling period, on days where participants reported any intake, 93% of daily summaries from DietMatePro records ($n = 80$) and 95% of paper diary ($n = 75$) days ($p = 0.37$) fell within range. Sensitivity analyses were performed to address the possibility that informative missing data may lead to incorrect inferences. Adherence was 78% for both conditions assuming drop-outs were completely non-compliant to recording ($p = 0.99$) and 95% for DietMatePro versus 96% for the paper diary under the assumption that all drop-outs were completely adherent to recording ($p = 0.30$).

Table 2. Effectiveness of Dietary Monitoring as Measured by Week 1 Median (IQR^a) and Spearman Correlations by Monitoring Condition

Energy or Nutrient	DietMatePro (n = 71)	Paper Diary (n = 78)	DietMatePro Correlations (n = 71)	Paper Diary Correlations (n = 78)
Energy (kcal) ^c	1872 (1376, 2511)	1671 (1344, 2173)	0.542	0.773
Total Fat (g)	60 (46, 96)	57 (41, 77)	0.610	0.669
Saturated Fat (g)	22 (13, 33)	19 (14, 30)	0.595	0.631
Cholesterol (mg)	193 (100, 342)	163 (83, 321)	0.705	0.807
Carbohydrate (g)	232 (168, 322)	214 (172, 266)	0.632	0.756
Fiber (g) ^c	17 (10, 25)	17 (13, 24)	0.549	0.765
Protein (g) ^c	79 (54, 97)	75 (50, 94)	0.409	0.751
Vitamin A (IU) ^{b,c}	3676 (1831, 8250)	4877 (2632, 9373)	0.495	0.779
Vitamin C (mg)	79 (28, 135)	72 (35, 111)	0.377	0.580
Calcium (mg) ^c	693 (459, 952)	734 (423, 1133)	0.436	0.831
Iron (mg) ^c	13 (9, 19)	12 (9, 17)	0.577	0.805

^a Inter-quartile range (25th %, 75th %), kcal = kilocalories g = grams, mg = milligrams.

^b Ln-transformed energy and nutrients compared by monitoring condition using unpaired t-test. Significant difference for vitamin A ($p = 0.025$).

^c Fisher-transformed correlations compared by unpaired t-test. Significant difference ($p < 0.05$) between monitoring conditions for energy, fiber, protein, vitamin A, calcium, and iron.

Adherence to the Dietary Intervention between Conditions

Daily adherence to the intervention diet was defined as reported intakes via assigned recording method below the prescribed calorie intake, <15% total calories from fat, <7% calories from saturated fat, and cholesterol intake below 200mg. Using these criteria, mean percent adherence to all four dietary criteria during the last three days of the intervention was 43% ($n = 72$) for the DietMatePro condition versus 28% ($n = 69$) for the paper diary condition ($\chi^2(1) p = 0.039$). Including missing data under the extreme assumptions that missing data indicated non-adherence reduced the difference between groups from 15% to 14% whereas assuming missing values indicated adherence to the dietary regimen reduced the difference further to 10% between groups.

Pre-post intervention reductions in energy intake, measured by 24-hr recall, were 2.1 times greater for DietMatePro compared to the paper diary condition ($p = 0.005$). Median reduction in total fat intake was 31g (IQR 8, 62) among DietMatePro versus 22g (IQR -13,45) among paper diary participants ($p = 0.032$).

Although the intervention period was too short to expect differential anthropometric changes from these dietary monitoring conditions, there were trends towards larger anthropometric changes in the DietMatePro versus paper diary condition over the 3-week intervention period. Mean weight decreases were 3.5 (SD = 4.9) pounds in the electronic compared to 2.9 (SD = 4.8) pounds in the paper diary condition ($p = 0.39$). Waist circumference decreased by 1.0 (SD = 1.2) inch in the DietMatePro compared to 0.5 (SD = 1.5) inch in the paper diary ($p = 0.04$). These findings were robust to adjustment for baseline BMI using linear regression models.

Self-Reported Adherence and Usability of the Paper-Based and DietMatePro Diaries

Though food record data suggested participants adhered to the prescribed dietary regimen for well under half of the meals, by questionnaire the DietMatePro diary group reported 60% adherence and the paper diary group reporting adhering to dietary recommendations for 63% of the meals. There were no significant differences between groups on 5-point Likert-Scale ratings of how closely the diet was followed, ease of following the diet, or utility of the recording method.

DISCUSSION

The results of this trial suggest that paper-based food diaries may be more concordant with 24-hr recall data than DietMatePro food diaries. The DietMatePro record was moderately correlated with the 24-hr dietary recall on calories and major macronutrients, but these correlations were generally lower than those found for the paper-based food diary. Since 24 hour recall data may be incomplete due to memory problems or the interview situation [25] and the DietMatePro food record entries reported higher intakes than recall reports on average, it is unclear whether the paper-based diary is more accurate than DietMatePro, or just more similar to potentially incomplete retrospective recalls.

A prior study of DietMatePro produced higher correlations with 24-hr recall data, i.e. 0.713 for energy compared to 0.542, than in the present study [11]. This prior study, however, involved participants who used DietMatePro for only a three day period, the last day of which was compared to 24-hr recall. Therefore, the additional recording burden prior to the 24-hr recall period (i.e. 7 days) and the anticipation of another three weeks of recording may have resulted in a less careful and

complete recording on the day corresponding to the 24-hr recall period. The percentage of days with missing or implausible food records was greater than twice as high among DietMatePro compared to paper diary data. One possible explanation for this difference is that it takes approximately 8 minutes to record each meal using the PDA program [11] which is presumably longer than it would take to use a paper food diary. If so, then it is important to note that computerized food diary recording methods are not immune from inaccuracies resulting from participant recording burden.

Although DietMatePro did not correlate as well with 24-hr recall as did the paper-based diary in this study, DietMatePro participants adhered better to the diet prescription than paper diary participants, maintaining intake under stringent energy, fat, saturated fat, and cholesterol restrictions 43% of the days sampled compared to 28% among the paper diary participants. Glanz et al. reported on a pilot behavioral intervention study among 33 participants of the Women's Health Initiative using PDA's to reduce fat intake to less than 20% of caloric intake and increase fruit and vegetable intake found that participants met fat and produce goals on 60% of days [15]. Therefore, PDA-based dietary monitoring and feedback programs appear to have a positive impact on adherence to low-fat diets.

Though both conditions received books and individual instructions to improve adherence, the DietMatePro program also included several features compared to the paper diary designed to improve adherence (e.g. calorie and nutrient feedback following recording of a meal, comparison of actual intake to target intake, preplanned recipes and meals consistent with diet). Future research could assess the relative importance of individualized feedback and meal plans, recipes that are readily available, and the platform for the food recording method in encouraging change in dietary intake.

Further support for increased adherence to the dietary regimen was provided by the 24-hr recall data showing a mean calorie pre-post difference of 490 calories in DietMatePro participants compared to 226 calories among paper-based diary participants. Mean reductions in fat intake were similar to the 10.9% decrease reported during the first year of the Women's Health Initiative intervention [26]. Although study duration was too short to expect differential changes in anthropometric measures, those in the DietMatePro condition experienced significantly greater reductions in waist circumference compared to those in the paper-based diary condition.

There are a number of study limitations that should be considered in interpreting the results of this study. The study participants were mostly women and were selected for access to health care and computer literacy that could limit the external validity of these findings. Sampling three days of food records before and after the intervention may not be an accurate reflection of adherence, as participants may have increased recording close to the study visits or decreased recording due to fatigue. Selection factors may have influenced study completion, as 18 DietMatePro compared to 7 paper diary participants

had implausible food record data for the concordance analysis. The concordance analysis was based on the final day of recording before the week 1 visit, so the difference between groups may be due to greater retrospective recording in the paper-based condition, not necessarily to poorer recording adherence in the DietMatePro condition. We performed sensitivity analysis under best and worst-case scenarios of adherence to insure findings were not due solely to selection bias.

Though we modeled 24-hr recall procedures after a five-step, multiple-pass method, this dietary assessment technique is subject to measurement error that increases random error in comparing the 24-hr recall data with food diary data. Furthermore, the 24-hr recalls may have been influenced by the prior recording since it is not typical for participants to prospectively record and also retrospectively report food intake for the same time period. The study design could have been improved by masking research assistants to treatment assignment for 24 hour recalls and other assessment procedures.

Other studies have investigated the use of PDA programs to improve adherence to complicated dietary regimens as a result of chronic disease. A six-month crossover trial designed to evaluate the effect of using a handheld self-monitoring program and found a mean HbA1C reduction of 0.825% after a 3-month intervention compared with control periods [27]. Another pilot study that tested the feasibility of using a PDA program among five individuals following a hemodialysis dietary regimen described trends of increased albumin, but larger studies are required to draw inferences [14]. These studies suggest PDA based programs can improve adherence to dietary regimens leading to improved clinical outcomes, and a study lasting 6 months suggested changes may be sustainable [13]. The utility of PDA-based dietary assessment software programs such as DietMatePro will be highly dependent upon both the accuracy and comprehensiveness of the food and nutrient database and the ease with which foods may be searched and selected by the end user.

This study provides evidence regarding the feasibility of using DietMatePro as a tool for improving short-term adherence to dietary regimens. Studies are needed to assess patterns of long-term use and impact of using this self-monitoring and dietary adherence tool on clinical outcomes.

CONCLUSION

DietMatePro food records may provide valid assessment of dietary intake. Though PDA programs require additional training compared to food diaries, the information returned does not need to be entered into a nutrient analysis program by another person, thereby improving time efficiency and reducing possibilities for error in transferring the information from paper to database. Immediate tailored feedback as well as other PDA program features may enhance adherence to dietary regimens compared to paper-based monitoring of food intake.

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